

2024 Bi H1 Q13

Section: Metabolism and Survival

Topic: Ectotherms and Thermoregulation (Adders)

Question Summary:

Adders are ectothermic snakes whose body temperature depends on the external environment. Which of the listed statements are correct?

Worked Solution:

- (1) Wide range of ecological niches: FALSE for ectotherms in general; their activity and distribution are constrained by external temperatures.
- (2) Use behavioural responses to maintain optimum metabolic rate: TRUE; basking in sun or seeking shade are classic behavioural thermoregulation strategies.
- (3) High energy costs to achieve homeostasis: FALSE; ectotherms have lower metabolic energy costs than endotherms.

Therefore, only statement 2 is correct.

Final Answer: **A** (2 only).

Revision Tips:

- Ectotherms rely on external heat sources -> low metabolic costs.
- Endotherms maintain constant body temperature metabolically -> high energy costs.

- Behavioural thermoregulation includes basking, shade seeking, and changing posture.